



# UNIT 05: CSS CODES

## Table of Contents

|                                            |    |
|--------------------------------------------|----|
| INTRODUCTION .....                         | 3  |
| LEARNING OBJECTIVES .....                  | 4  |
| 5.1 CSS CODES: CODE STRUCTURE .....        | 4  |
| 5.1.1 CLASSICAL LINEAR CODES REVIEW .....  | 4  |
| 5.1.2 THE CSS CONSTRUCTION .....           | 5  |
| 5.1.3 THE STEANE CODE $[[7, 1, 3]]$ .....  | 7  |
| 5.2 SYNDROME MEASUREMENT .....             | 9  |
| 5.2.1 STABILISER MEASUREMENT BASICS .....  | 9  |
| 5.2.2 CSS SYNDROME STRUCTURE .....         | 9  |
| 5.2.3 SYNDROME EXTRACTION CIRCUITS.....    | 10 |
| 5.3 ERROR CORRECTION.....                  | 11 |
| 5.3.1 CSS ERROR CORRECTION PROCEDURE ..... | 11 |
| 5.3.2 HANDLING CONTINUOUS ERRORS .....     | 13 |
| 5.3.3 ADVANTAGES OF CSS STRUCTURE .....    | 14 |
| 5.4 SUMMARY .....                          | 15 |
| 5.5 MIND MAP.....                          | 16 |
| 5.6 GLOSSARY .....                         | 17 |
| 5.7 SELF-ASSESSMENT QUESTIONS.....         | 17 |
| 5.8 TERMINAL QUESTIONS.....                | 21 |



|                                      |    |
|--------------------------------------|----|
| 5.8.1 SHORT ANSWER QUESTIONS.....    | 21 |
| 5.8.2 LONG ANSWER QUESTIONS .....    | 21 |
| 5.9 ANSWERS .....                    | 22 |
| 5.9.1 SELF-ASSESSMENT ANSWERS.....   | 22 |
| 5.9.2 TERMINAL QUESTION ANSWERS..... | 23 |
| 5.10 REFERENCES .....                | 24 |



## INTRODUCTION

In Unit 4, we explored Shor's 9-qubit code — the first quantum error-correcting code capable of protecting against arbitrary single-qubit errors. While groundbreaking, Shor's approach uses concatenation: three groups of three qubits, requiring significant overhead. A natural question arises: can we do better? Can the rich theory of classical error-correcting codes, developed over decades, be directly harnessed to protect quantum information?

The answer came in 1996 through independent discoveries by Robert Calderbank and Peter Shor, and separately by Andrew Steane. They found a systematic method to construct quantum codes directly from classical linear codes — a construction now known as CSS codes after the initials of its inventors. This breakthrough demonstrated that classical coding theory, which underlies everything from CDs to satellite communications, contains the mathematical machinery needed for quantum error protection.

CSS codes represent more than a theoretical curiosity. The Steane code, a CSS code encoding one logical qubit into seven physical qubits, has been implemented in experimental quantum computers by Honeywell/Quantinuum, the University of Innsbruck, and other research groups. Understanding CSS construction is essential for anyone working in quantum computing because these codes form the foundation of many important quantum error correction schemes, including the surface code, which is the leading practical scheme today.

The key insight behind CSS codes is that quantum errors can be decomposed into bit-flip (X) and phase-flip (Z) components. By using two carefully chosen classical codes — one to correct bit-flips and another to correct phase-flips — we can construct a quantum code that handles both error types. The requirement that these classical codes satisfy a specific dual code constraint ensures that the resulting quantum code has well-defined stabiliser generators that commute with each other.



This unit builds on the Knill-Laflamme conditions from Unit 3 and the concatenation approach from Unit 4. We will see how CSS codes satisfy the error correction conditions systematically through their construction. In Unit 6, we will generalise the CSS framework to the full stabiliser formalism, which encompasses all codes where generators are products of Pauli operators.

## LEARNING OBJECTIVES

By the end of this unit, you will be able to:

1. Construct CSS codes from classical linear codes satisfying the dual code constraint.
2. Analyse the stabiliser structure of CSS codes and its role in error detection.
3. Implement syndrome measurement and error correction for CSS codes.
4. Evaluate the Steane  $[[7, 1, 3]]$  code as a practical CSS implementation.
5. Compare CSS codes with general quantum codes for fault-tolerant computation.

## 5.1 CSS CODES: CODE STRUCTURE

### 5.1.1 CLASSICAL LINEAR CODES REVIEW

Before constructing CSS codes, we must understand the classical codes that serve as their building blocks. A classical linear code  $C$  is a subspace of the vector space  $F_2^n$  (binary vectors of length  $n$ ). Two matrices characterise any linear code: the generator matrix  $G$  and the parity check matrix  $H$ .

The **generator matrix**  $G$  is a  $k \times n$  matrix whose rows form a basis for the code. Any codeword  $c$  can be written as  $c = xG$  for some information vector  $x$  of length  $k$ . The code thus contains  $2^k$  codewords.

The **parity check matrix**  $H$  is an  $(n - k) \times n$  matrix satisfying  $Hc = 0$  for all codewords  $c$ . This matrix detects errors: if we receive  $y = c + e$  (codeword plus error), then  $Hy =$



$Hc + He = He$ . The syndrome  $s = He$  depends only on the error, not on the transmitted codeword.

These matrices satisfy the fundamental relationship  $GH^T = 0$ , meaning every row of  $G$  is orthogonal to every row of  $H$ .

The **dual code**  $C^\perp$  consists of all vectors orthogonal to every codeword in  $C$ :

$$C^\perp = \{y \in F_2^n : y \cdot c = 0 \text{ for all } c \in C\}$$

**How to read:** "C-perp equals the set of all  $y$  in  $F_2^n$  such that  $y \cdot c$  equals zero for all  $c$  in  $C$ "

If  $C$  has generator matrix  $G$  and parity check matrix  $H$ , then  $C^\perp$  has generator matrix  $H^T$  and parity check matrix  $G^T$ . The dimensions satisfy: if  $C$  is an  $[n, k]$  code, then  $C^\perp$  is an  $[n, n - k]$  code.

### Example: The [7, 4, 3] Hamming Code

The classical Hamming code is a  $[7, 4, 3]$  code: 7 bits total, 4 information bits, and minimum distance 3 (can correct any single-bit error). Its parity check matrix is:

$$H = [0001111 \ 0110011 \ 1010101]$$

Each column of  $H$  is the binary representation of the numbers 1 through 7 (reading top to bottom as least to most significant bit). If error  $e$  is a single-bit flip at position  $j$ , then  $He$  equals the  $j$ -th column of  $H$ , directly identifying the error location.

#### *Think About It*

*In the  $[7, 4, 3]$  Hamming code, the syndrome  $He$  uniquely identifies single-bit errors because each column of  $H$  is distinct and non-zero. With 3 syndrome bits, we can distinguish  $2^3 - 1 = 7$  non-trivial syndromes, exactly matching the 7 possible single-bit error positions.*

## 5.1.2 THE CSS CONSTRUCTION

The CSS construction builds a quantum code from two classical linear codes. Let  $C_1$  be an  $[n, k_1]$  code and  $C_2$  be an  $[n, k_2]$  code, both with the same block length  $n$ .

### The Dual Code Constraint



For the CSS construction to produce a valid quantum code, we require:

$$C_2 \subset C_1 (\text{equivalently, } C_1^\perp \subseteq C_2)$$

This constraint ensures that X-type and Z-type stabiliser generators commute. Without it, some stabilisers would anti-commute, making simultaneous measurement impossible and invalidating the code.

The mathematical reason is as follows: X-type generators are derived from the parity check matrix  $H_1$  of  $C_1$ , while Z-type generators come from  $H_2$  of  $C_2$ . Two Pauli operators commute if they share an even number of positions where both are non-trivial. For rows  $r_1$  of  $H_1$  and  $r_2$  of  $H_2$ , the corresponding stabilisers commute if and only if  $r_1 \cdot r_2 = 0$ . This happens for all such pairs precisely when  $C_1^\perp \subseteq C_2$ .

### CSS Codeword States

Given codes satisfying  $C_2 \subset C_1$ , the CSS codewords are uniform superpositions over cosets of  $C_2$  in  $C_1$ :

$$|x + C_2\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} |x + y\rangle$$

**How to read:** "The state ket  $x$  plus  $C_2$  equals one over the square root of the size of  $C_2$ , times the sum over all  $y$  in  $C_2$  of ket  $x$  plus  $y$ "

**Where:**

- $|x + C_2\rangle$  is the quantum codeword corresponding to coset  $x + C_2$
- $|C_2| = 2^{k_2}$  is the number of codewords in  $C_2$
- $x + y$  denotes bitwise XOR (addition modulo 2)
- The sum is over all  $2^{k_2}$  codewords  $y$  in  $C_2$

The **dimension** of the CSS code equals the number of distinct cosets of  $C_2$  in  $C_1$ :

$$\dim(\text{CSS}(C_1, C_2)) = \frac{|C_1|}{|C_2|} = 2^{k_1 - k_2}$$

Thus  $\text{CSS}(C_1, C_2)$  is an  $[[n, k_1 - k_2]]$  quantum code, encoding  $k_1 - k_2$  logical qubits into  $n$  physical qubits.

*Common Misconception*



"CSS codes require  $C_1 = C_2$ " — FALSE. The Steane code uses  $C_1 = C$  (Hamming code) and  $C_2 = C^\perp$  (dual Hamming code). The requirement is  $C_2 \subset C_1$ , not equality. Many interesting CSS codes use different codes for  $C_1$  and  $C_2$ .

### Error Correction Capability

The CSS code corrects  $t$  errors if both  $C_1$  and  $C_2^\perp$  can correct  $t$  errors. Specifically,  $C_1$  must correct bit-flip errors (through Z-type syndrome measurement), and  $C_2^\perp$  must correct phase-flip errors (through X-type syndrome measurement).

### 5.1.3 THE STEANE CODE $[[7, 1, 3]]$

The Steane code is the primary example of a CSS code and demonstrates the construction beautifully. It uses:

- $C_1 = C$ : The  $[7,4,3]$  Hamming code
- $C_2 = C^\perp$ : The dual  $[7,3,4]$  Hamming code

The Hamming code has a special property: it is **weakly self-dual**, meaning  $C^\perp \subset C$ . This satisfies the CSS constraint, so the construction is valid. The resulting quantum code has parameters:

$$[[n, k_1 - k_2, d]] = [[7, 4 - 3, 3]] = [[7, 1, 3]]$$

This encodes 1 logical qubit into 7 physical qubits and corrects any single-qubit error.

### Logical Codewords

The logical  $|0_L\rangle$  is a superposition over all 8 codewords in  $C_2 = C^\perp$ :

$$|0_L\rangle = \frac{1}{\sqrt{8}} [|0000000\rangle + |1010101\rangle + |0110011\rangle + |1100110\rangle + |0001111\rangle \\ + |1011010\rangle + |0111100\rangle + |1101001\rangle]$$

The logical  $|1_L\rangle$  corresponds to the other coset (obtained by adding any codeword in  $C_1 \setminus C_2$ ):



$$|1_L\rangle = \frac{1}{\sqrt{8}} [|1111111\rangle + |0101010\rangle + |1001100\rangle + |0011001\rangle + |1110000\rangle \\ + |0100101\rangle + |1000011\rangle + |0010110\rangle]$$

Notice that  $|1_L\rangle = \bar{X}|0_L\rangle$  where  $\bar{X} = X_1X_2X_3X_4X_5X_6X_7$  is the logical X operator (all X gates applied to all qubits).

### Stabiliser Generators

The Steane code has 6 stabiliser generators: 3 X-type and 3 Z-type. These correspond directly to the rows of the Hamming code's parity check matrix  $H$ :

| Generator | Operator | Type   | Pattern           |
|-----------|----------|--------|-------------------|
| $g_1$     | IIIXXXX  | X-type | Positions 4,5,6,7 |
| $g_2$     | IXXIIXX  | X-type | Positions 2,3,6,7 |
| $g_3$     | XIXIXIX  | X-type | Positions 1,3,5,7 |
| $g_4$     | IIIZZZZ  | Z-type | Positions 4,5,6,7 |
| $g_5$     | IZZIIZZ  | Z-type | Positions 2,3,6,7 |
| $g_6$     | ZIZIZIZ  | Z-type | Positions 1,3,5,7 |

**Table 5.1: Stabiliser Generators of the Steane  $[[7, 1, 3]]$  Code — Three X-type and Three Z-type Generators Derived from the Hamming Code Parity Check Matrix.**

The support patterns match the rows of  $H$ : row 1 has 1s in positions 4,5,6,7, and so on. X-type generators use X where  $H$  has 1, and I elsewhere; similarly for Z-type.

### Logical Operators

The logical operators for the Steane code are:

$$\bar{Z} = Z_1Z_2Z_3Z_4Z_5Z_6Z_7$$

$$\bar{X} = X_1X_2X_3X_4X_5X_6X_7$$

These satisfy  $\bar{Z}|0_L\rangle = +|0_L\rangle$ ,  $\bar{Z}|1_L\rangle = -|1_L\rangle$ , and  $\bar{X}|0_L\rangle = |1_L\rangle$ ,  $\bar{X}|1_L\rangle = |0_L\rangle$ , as required for logical Z and X operations.

*Did You Know?*





*The Steane code was one of the first quantum codes implemented experimentally. In 2021, researchers at Honeywell demonstrated logical qubit operations using the Steane code in a trapped-ion quantum computer, achieving error rates below the code's threshold.*

## 5.2 SYNDROME MEASUREMENT

### 5.2.1 STABILISER MEASUREMENT BASICS

Syndrome measurement in stabiliser codes involves measuring the eigenvalue of each stabiliser generator. Since all generators are products of Pauli operators, their eigenvalues are  $\pm 1$ . We encode these as syndrome bits:

- Eigenvalue  $+1 \rightarrow$  syndrome bit 0
- Eigenvalue  $-1 \rightarrow$  syndrome bit 1

For an  $[[n, k]]$  stabiliser code with  $n - k$  generators, we obtain  $n - k$  syndrome bits. For the Steane code  $[[7, 1, 3]]$ , this gives 6 syndrome bits.

The critical property of syndrome measurement is that it reveals error information without collapsing the logical superposition. When we measure stabiliser  $P$  on state  $E|\psi\rangle_L$  (encoded state with error  $E$ ), the result depends on whether  $E$  commutes or anti-commutes with  $P$ :

$$E|\psi\rangle_L|0\rangle_A \rightarrow \frac{1}{2}(1 + P)E|\psi\rangle_L|0\rangle_A + \frac{1}{2}(1 - P)E|\psi\rangle_L|1\rangle_A$$

**How to read:** "After syndrome extraction, if stabiliser  $P$  commutes with error  $E$ , the ancilla is 0; if  $P$  anti-commutes with  $E$ , the ancilla is 1"

After measuring the ancilla:

- If  $[E, P] = 0$  (commute): measurement yields 0, state remains in  $+1$  eigenspace
- If  $\{E, P\} = 0$  (anti-commute): measurement yields 1, state is in  $-1$  eigenspace



### 5.2.2 CSS SYNDROME STRUCTURE

CSS codes have a special structure that simplifies syndrome measurement: X-type and Z-type stabilisers detect different error types.

**X-type stabilisers detect Z errors:** An X-type stabiliser (product of X and I only) anti-commutes with Z operators at matching positions. Recall that  $XZ = -ZX$ . So if stabiliser  $g$  has X on qubit  $j$  and error  $E$  has Z on qubit  $j$ , they anti-commute.

**Z-type stabilisers detect X errors:** Similarly, Z-type stabilisers anti-commute with X operators at matching positions.

| Error Type                       | Anti-commutes With                  | Detected By    |
|----------------------------------|-------------------------------------|----------------|
| $X_j$ (bit-flip on qubit $j$ )   | Z-type stabilisers containing $Z_j$ | Z syndrome     |
| $Z_j$ (phase-flip on qubit $j$ ) | X-type stabilisers containing $X_j$ | X syndrome     |
| $Y_j = iX_jZ_j$                  | Both types                          | Both syndromes |

**Table 5.2: Error Detection Structure in CSS Codes — X-type Stabilisers Detect Z Errors and Z-type Stabilisers Detect X Errors, Enabling Independent Correction.**

This separation is the key advantage of CSS codes: we can independently detect and correct X and Z errors.

#### **Common Misconception**

*"The syndrome measurement reveals the encoded quantum state" — FALSE. Syndrome measurement extracts only error information. The logical superposition  $\alpha|0_L\rangle + \beta|1_L\rangle$  remains intact because stabiliser measurements project onto error subspaces, not onto computational basis states.*



### 5.2.3 SYNDROME EXTRACTION CIRCUITS

#### Z-type Stabiliser Measurement

To measure a Z-type stabiliser like  $g_4 = IIIZZZZ$  (acting on qubits 4,5,6,7):

1. Prepare ancilla qubit in  $|0\rangle$
2. Apply CNOT from qubit 4 to ancilla
3. Apply CNOT from qubit 5 to ancilla
4. Apply CNOT from qubit 6 to ancilla
5. Apply CNOT from qubit 7 to ancilla
6. Measure ancilla in computational basis

The ancilla computes the parity (XOR) of qubits 4-7. If the parity is even, result is 0 (+1 eigenvalue); if odd, result is 1 (−1 eigenvalue).

#### X-type Stabiliser Measurement

For X-type stabilisers like  $g_1 = IIIXXXX$ , we use the Hadamard transformation:

7. Apply Hadamard to qubits 4,5,6,7
8. Apply the Z-type measurement circuit above
9. Apply Hadamard to qubits 4,5,6,7 again

The Hadamard gates transform  $X \leftrightarrow Z$ , converting the X-type measurement into a Z-type measurement, then returning to the original basis.

#### Worked Example: Measuring $g_4 = IIIZZZZ$

Consider the Steane code with an  $X_5$  error (bit-flip on qubit 5). Before error: logical state  $|\psi_L\rangle$ . After error:  $X_5|\psi_L\rangle$ .

When we measure  $g_4 = IIIZZZZ$ :

- $g_4$  acts on qubits 4,5,6,7 with Z operators
- $X_5$  anti-commutes with the  $Z_5$  in  $g_4$
- Therefore, measurement yields −1 (syndrome bit 1)

This non-trivial syndrome indicates an error affecting the parity of qubits 4-7.

#### *Tips and Tricks*



*For CSS codes, you can measure all Z-type stabilisers first, then all X-type stabilisers (or vice versa). The order does not matter because X-type and Z-type stabilisers commute with each other, and measuring one type does not disturb the outcomes of the other.*

## 5.3 ERROR CORRECTION

### 5.3.1 CSS ERROR CORRECTION PROCEDURE

The CSS error correction procedure exploits the separated stabiliser structure to independently correct X and Z errors. Here is the complete algorithm:

#### Step 1: Measure Z-type stabilisers

Measure all Z-type generators ( $g_4, g_5, g_6$  for Steane code) to obtain the X-error syndrome  $s_X$ . This syndrome equals  $H_2 e_1$  where  $e_1$  is the X-error pattern and  $H_2$  is the parity check matrix of  $C_2$ .

#### Step 2: Classical decoding for X errors

Use the syndrome  $s_X$  to identify the X-error pattern  $e_1$ . This uses the same classical decoding algorithm used for  $C_2$ . For the Steane code (using Hamming codes), the syndrome directly indicates the error position.

#### Step 3: Apply X corrections

Apply  $X_j$  for each position  $j$  where  $e_1$  has a 1. This removes all bit-flip errors.

#### Step 4: Measure X-type stabilisers

Measure all X-type generators ( $g_1, g_2, g_3$  for Steane code) to obtain the Z-error syndrome  $s_Z$ . This syndrome equals  $H_1 e_2$  where  $e_2$  is the Z-error pattern.

#### Step 5: Apply Z corrections

Apply  $Z_j$  for each position  $j$  where  $e_2$  has a 1. This removes all phase-flip errors.

**Result:** The original encoded state is recovered.

**Worked Example: Correcting  $X_3$  Error on Steane Code**



Initial state:  $|\psi_L\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$

Error:  $X_3$  (bit-flip on qubit 3)

Corrupted state:  $X_3|\psi_L\rangle$

**Syndrome measurement (Z-type stabilisers):**

- Measure  $g_4 = IIIZZZZ$ : qubit 3 not in support, eigenvalue +1, syndrome bit 0
- Measure  $g_5 = IZZIIZZ$ : qubit 3 has Z,  $X_3$  anti-commutes, eigenvalue -1, syndrome bit 1
- Measure  $g_6 = ZIZIZIZ$ : qubit 3 has Z,  $X_3$  anti-commutes, eigenvalue -1, syndrome bit 1

Syndrome = (0,1,1)

**Decoding:** In the Hamming code, syndrome (0,1,1) corresponds to column 3 of  $H$ , identifying position 3.

**Correction:** Apply  $X_3$

**Recovered state:**  $X_3 \cdot X_3|\psi_L\rangle = |\psi_L\rangle \checkmark$

The X-type syndrome measurement would give all 0s (no Z errors present), requiring no further correction.

### 5.3.2 HANDLING CONTINUOUS ERRORS

A remarkable property of quantum error correction is that it handles continuous errors despite using discrete corrections. Any single-qubit error operator can be written as:

$$O = aI + bX + cY + dZ, \text{ where } a,b,c,d \in \mathbb{C}$$

for some complex coefficients  $a, b, c, d$ . When syndrome measurement occurs, the quantum state is projected onto one of the discrete error subspaces:

1. If the state was in the code space (I error), it remains there
2. If an X error occurred, the state is projected to the X-error subspace
3. Similarly for Y and Z errors



After projection, the standard Pauli correction restores the state to the code space. The net effect: continuous errors are discretised by measurement and then corrected.

### Success Probability for Steane Code

The Steane code succeeds if at most one qubit experiences an error:

$$P_{\text{success}} = (1 - p)^7 + 7p(1 - p)^6$$

**How to read:** "Probability of success equals probability of no error on all 7 qubits, plus probability of exactly one error (which we can correct)"

**Where:**

- $p$  = single-qubit error probability
- $(1 - p)^7$  = probability all 7 qubits are error-free
- $7p(1 - p)^6$  = probability exactly one of 7 qubits has an error

**Benefit threshold:** The code provides improvement when  $P_{\text{success}} > 1 - p$ . For the simplified model above (independent errors, perfect syndrome measurement), this gives a pseudo-threshold of approximately  $p \lesssim 0.0579$  (5.79%). However, when accounting for errors in syndrome extraction circuits (fault-tolerant threshold), the threshold is significantly lower, typically in the range of 0.1–1% depending on implementation details.

#### *Did You Know?*

*The pseudo-threshold of 5.79% applies to an idealised model. In practice, the fault-tolerant threshold (accounting for imperfect syndrome extraction) is lower. Modern superconducting qubits typically achieve 0.1–1% two-qubit gate error rates, while trapped-ion qubits can achieve even lower two-qubit gate error rates, around 0.01%.*

### 5.3.3 ADVANTAGES OF CSS STRUCTURE

CSS codes offer several advantages over general quantum error-correcting codes:

#### Independent X and Z correction



The separated stabiliser structure means X errors and Z errors can be diagnosed and corrected independently. This simplifies both the syndrome measurement circuits and the classical decoding logic.

### Classical decoding algorithms

The syndrome-to-error mapping for CSS codes builds on classical coding theory. Research on classical decoders for Hamming codes, BCH codes, and LDPC codes provides a starting point for CSS code decoders, though adaptations are required to handle quantum-specific challenges such as code degeneracy and stabiliser commutativity constraints.

### Transversal gates

A transversal gate is one where each physical qubit interacts only with the corresponding qubit in another code block:

As established by Gottesman and others, transversal operations — where each qubit in a block interacts only with the corresponding qubit in another block or a specialised ancilla — are inherently fault-tolerant.

For CSS codes:

- **CNOT** is transversal for all CSS codes: apply CNOT from qubit  $i$  of one block to qubit  $i$  of another block, for all  $i$
- **Hadamard** is transversal for self-dual CSS codes (where  $C_1 = C_2^\perp$ , like the Steane code): apply H to each qubit independently

Transversal gates prevent error propagation within a code block, making them essential for fault-tolerant quantum computation.

### Systematic construction

Given any classical code family satisfying the dual code constraint, we can systematically construct corresponding CSS codes. This provides a rich library of quantum codes with known properties.

### *Activity*



Consider the classical  $[15,11,3]$  Hamming code and its dual  $[15,4,8]$  code. Verify that  $C^\perp \subset C$  (the dual code constraint is satisfied). Then determine the parameters  $[[n,k,d]]$  of the resulting CSS code and calculate how many logical qubits it encodes.

## 5.4 SUMMARY

In this unit, you learnt about CSS codes, a systematic construction for quantum error-correcting codes derived from pairs of classical linear codes. The key concepts are:

- **CSS construction** requires two classical codes  $C_1$  and  $C_2$  satisfying the dual code constraint  $C_2 \subset C_1$  (equivalently,  $C_1^\perp \subseteq C_2$ )
- **CSS codewords** are uniform superpositions over cosets:  $|x + C_2\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} |x + y\rangle$
- The **dimension** of  $\text{CSS}(C_1, C_2)$  equals  $2^{k_1 - k_2}$ , encoding  $k_1 - k_2$  logical qubits into  $n$  physical qubits
- The **Steane  $[[7, 1, 3]]$  code** is built from the  $[7,4,3]$  Hamming code and its dual, encoding one logical qubit into seven physical qubits
- Steane code has **6 stabiliser generators**: 3 X-type (from rows of  $H$ ) and 3 Z-type (same pattern)
- **X-type stabilisers detect Z errors** because  $X_i$  and  $Z_i$  anti-commute; **Z-type stabilisers detect X errors** for the same reason
- **Syndrome measurement** is non-destructive: it reveals error information while preserving the encoded superposition
- **CSS error correction** proceeds in two phases: first correct X errors using Z-syndrome, then correct Z errors using X-syndrome (or vice versa)
- **Continuous errors are discretised** by syndrome measurement, allowing correction of arbitrary single-qubit errors using discrete Pauli corrections





- The Steane code **pseudo-threshold** (idealised model) is approximately  $p \lesssim 5.79\%$ ; the **fault-tolerant threshold** accounting for circuit-level noise is lower (typically 0.1–1%)
- **Transversal CNOT** is available for all CSS codes; **transversal Hadamard** is available for self-dual CSS codes like Steane
- **Transversal gates are automatically fault-tolerant** because errors cannot propagate within a code block
- **Classical decoding algorithms** provide a foundation for CSS syndrome decoding, though adaptations are needed for quantum-specific challenges
- CSS codes form the **foundation for fault-tolerant quantum computation** and are implemented in current experimental quantum computers

## 5.5 MIND MAP



[This page is reserved for the unit mind map]

## 5.6 GLOSSARY

**Coset** — A partition subset  $x + C_2$  of vectors sharing the same syndrome in a parent code.

**CSS Code** — Quantum code built from two classical linear codes with separated X-type and Z-type stabilisers.

**Dual Code** — The set  $C^\perp$  of all vectors orthogonal to every codeword in a linear code  $C$ .

**Parity Check Matrix** — Matrix  $H$  satisfying  $Hc = 0$  for all codewords; used for syndrome-based error detection.

**Steane Code** — The  $[[7, 1, 3]]$  CSS code from the Hamming code; corrects any single-qubit error.

**Syndrome** — Stabiliser measurement outcome identifying the type and location of errors.

**Transversal Operation** — Gate where each qubit interacts only with its counterpart in another block; inherently fault-tolerant.

**X-type Stabiliser** — Generator containing only Pauli X and identity; detects Z (phase-flip) errors.

**Z-type Stabiliser** — Generator containing only Pauli Z and identity; detects X (bit-flip) errors.

## 5.7 SELF-ASSESSMENT QUESTIONS

1. Why do X-type stabilisers detect Z errors but not X errors?
  - a) X and Z operators always commute
  - b) X and Z operators anti-commute when acting on the same qubit
  - c) X-type stabilisers contain Z operators
  - d) Z errors do not affect X-type stabilisers



2. Two CSS codewords  $|x + C_2\rangle$  and  $|x' + C_2\rangle$  are considered the same logical state whenever  $x$  and  $x'$  differ by a word in  $C_2$ . Why does using bitwise XOR (mod-2 addition) in the coset  $x + C_2$  ensure that these codewords are indistinguishable by Z-type stabiliser measurements?
  - a) Because XOR produces bit strings of the same length, making all codewords look identical
  - b) Because the Z-type stabilisers correspond to generators of  $C_2$ , so any two strings in the same coset yield the same Z-parity outcomes
  - c) Because mod-2 addition preserves the Hamming weight of every codeword
  - d) Because XOR is commutative, ensuring measurement order does not matter
3. Why must  $C_2 \subset C_1$  for a valid CSS construction?
  - a) To ensure the quantum code has positive dimension
  - b) To ensure X-type and Z-type stabilisers commute
  - c) To maximise the number of encoded qubits
  - d) To enable transversal Hadamard gates
4. How does syndrome measurement preserve the encoded quantum state?
  - a) It measures in a basis orthogonal to the code space
  - b) It projects onto error subspaces, not onto basis states of the logical information
  - c) It uses non-demolition measurement techniques
  - d) It only measures ancilla qubits, never data qubits
5. For the Steane code, if measuring  $g_4 = IIIZZZZ$ ,  $g_5 = IZZIIZZ$ , and  $g_6 = ZIZIZIZ$  gives syndrome  $(1,1,0)$ , which qubit has an X error?
  - a) Qubit 2
  - b) Qubit 4
  - c) Qubit 5
  - d) Qubit 6



6. Given classical codes  $C_1 = [15,11]$  and  $C_2 = [15,4]$  with  $C_2 \subset C_1$ , how many logical qubits does  $\text{CSS}(C_1, C_2)$  encode?
- a) 4 qubits
  - b) 7 qubits
  - c) 11 qubits
  - d) 15 qubits
7. The Z-type stabiliser  $IIIZZZZ$  can be measured directly using CNOT gates with data qubits as controls. However, the X-type stabiliser  $IIIXXXX$  cannot be measured with the same CNOT circuit alone. Why are Hadamard gates on qubits 4-7 required before and after the CNOT circuit to measure  $IIIXXXX$ ?
- a) Hadamard gates entangle the ancilla with the data qubits, which is necessary for X-type measurement
  - b) Hadamard gates convert X operators to Z operators and back, allowing the Z-parity CNOT circuit to effectively measure X-parity
  - c) Hadamard gates reduce the circuit depth, making the measurement faster
  - d) Hadamard gates are not strictly required; they merely improve measurement fidelity
8. How do the syndrome extraction circuits for X-type and Z-type stabilisers differ?
- a) X-type uses more ancilla qubits
  - b) Z-type requires Hadamard transformation while X-type does not
  - c) X-type requires Hadamard transformation while Z-type does not
  - d) They use completely different gate sets with no common elements
9. The Steane code benefit threshold is approximately 5.79%. Which statement best explains this threshold?
- a) Above this error rate, the code space becomes undefined
  - b) Above this error rate, two-qubit errors become more likely than one-qubit errors, which the code cannot correct



- c) Above this error rate, the syndrome measurement itself introduces more errors than it corrects
  - d) Above this error rate, transversal gates cease to function
10. Comparing the Steane  $[[7, 1, 3]]$  code to Shor's  $[[9, 1, 3]]$  code, which structural difference most directly enables transversal Hadamard in Steane but not Shor?
- a) Steane uses fewer physical qubits
  - b) Steane's underlying classical code is self-dual
  - c) Steane has separated X-type and Z-type stabilisers
  - d) Shor uses concatenation while Steane does not
11. The number of logical qubits encoded by  $\text{CSS}(C_1, C_2)$  where  $C_1$  is an  $[n, k_1]$  code and  $C_2$  is an  $[n, k_2]$  code equals .
- a)  $k_1 + k_2$
  - b)  $k_1 - k_2$
  - c)  $n - k_1$
  - d)  $n - k_2$
12. In the Steane code, measuring Z-type stabilisers produces syndrome bits that identify the location of errors.
- a) Z (phase-flip)
  - b) X (bit-flip)
  - c) Y (combined)
  - d) Continuous
13. True or false: In a CSS code, X-type and Z-type stabiliser generators can anti-commute if the dual code constraint is satisfied.
- a) True, because X and Z operators always anti-commute
  - b) False, the dual code constraint ensures all generators commute
  - c) True, but only when both generators act on the same qubit
  - d) False, because CSS codes do not use Z-type stabilisers



14. True or false: The Steane code can correct arbitrary two-qubit errors because it has distance 3.
- a) True, distance 3 means it corrects up to 3 errors
  - b) True, the 7 physical qubits provide enough redundancy for two-qubit correction
  - c) False, distance 3 allows correction of at most one arbitrary error
  - d) False, the Steane code cannot correct any errors

## 5.8 TERMINAL QUESTIONS

### 5.8.1 SHORT ANSWER QUESTIONS

1. Explain why the dual code constraint  $C_2 \subset C_1$  is necessary for CSS construction. What happens if this constraint is violated?
2. Describe the syndrome extraction circuit for measuring a Z-type stabiliser like  $IIIZZ$ . What role does the ancilla qubit play?
3. What happens to a continuous single-qubit error ( $O = aI + bX + cY + dZ$ ) during CSS error correction? How does measurement discretise the error?
4. Why are transversal gates automatically fault-tolerant? Give an example from the Steane code.
5. How does the Steane code's stabiliser structure enable independent correction of X and Z errors? Why is this separation advantageous?

### 5.8.2 LONG ANSWER QUESTIONS

1. Explain the relationship between the Hamming code's parity check matrix  $H$  and the stabiliser generators of the Steane code. How are the X-type and Z-type generators derived from  $H$ ?



2. Compare the Steane  $[[7, 1, 3]]$  code with Shor's  $[[9, 1, 3]]$  code. Discuss their construction methods, resource overhead, stabiliser structure, and which transversal gates each supports.
3. Design a complete error correction procedure for the Steane code when a  $Y_5$  error occurs. Show the syndrome measurement results for both Z-type and X-type stabilisers, identify the error, and describe the correction steps.
4. Analyse why CSS codes enable simpler fault-tolerant gate implementations compared to general stabiliser codes. Discuss the role of the separated X-type and Z-type structure.
5. A quantum engineer needs to choose between the Steane code and a hypothetical  $[[11, 1, 5]]$  code for a system with 2% physical error rate. The  $[[11, 1, 5]]$  code can correct up to 2 errors but has no transversal Hadamard. Evaluate the trade-offs and recommend a choice.

## 5.9 ANSWERS

### 5.9.1 SELF-ASSESSMENT ANSWERS

1. b) X and Z operators anti-commute when acting on the same qubit
2. b) Because the Z-type stabilisers correspond to generators of  $C_2$ , so any two strings in the same coset yield the same Z-parity outcomes
3. b) To ensure X-type and Z-type stabilisers commute
4. b) It projects onto error subspaces, not onto basis states of the logical information
5. d) Qubit 6
6. b) 7 qubits
7. b) Hadamard gates convert X operators to Z operators and back, allowing the Z-parity CNOT circuit to effectively measure X-parity
8. c) X-type requires Hadamard transformation while Z-type does not



9. b) Above this error rate, two-qubit errors become more likely than one-qubit errors, which the code cannot correct
10. b) Steane's underlying classical code is self-dual
11. b)  $k_1 - k_2$
12. b) X (bit-flip)
13. b) False, the dual code constraint ensures all generators commute
14. c) False, distance 3 allows correction of at most one arbitrary error

## 5.9.2 TERMINAL QUESTION ANSWERS

### SHORT ANSWER QUESTIONS

1. The constraint  $C_2 \subset C_1$  ensures X-type and Z-type stabiliser generators commute by guaranteeing  $r_1 \cdot r_2 = 0$  for all generator pairs. Violation causes anti-commutation, making simultaneous measurement impossible and invalidating the code.
2. Prepare an ancilla in  $|0\rangle$ , apply CNOT from each data qubit in the stabiliser's support to the ancilla. The ancilla accumulates parity; measuring it yields the syndrome bit without disturbing the encoded state.
3. Syndrome measurement projects the continuous error  $O = aI + bX + cY + dZ$  onto a discrete Pauli subspace. After projection, standard Pauli correction restores the code space. Measurement discretises continuous errors into correctable Pauli types.
4. Transversal gates act independently on each qubit, preventing single-qubit errors from propagating within a code block. In the Steane code, transversal CNOT applies qubit-by-qubit between blocks, keeping errors within correction capability.
5. Three X-type generators detect Z errors and three Z-type generators detect X errors independently. Each type uses classical Hamming decoding. The two stabiliser types commute, so measurement order is flexible and correction phases are independent.

### LONG ANSWER QUESTIONS





1. Each row of the Hamming code's  $3 \times 7$  parity check matrix  $H$  defines a support pattern. X-type generators place X operators where  $H$  has 1s and identity elsewhere — for example, row (0001111) yields  $g_1 = IIIXXXX$ . Z-type generators use the same patterns with Z operators, giving  $g_4 = IIIZZZZ$  from the same row. All six generators share three support patterns from  $H$ , with three using X and three using Z. Refer to section 5.1.3 for the complete generator table.
2. Shor's  $[[9, 1, 3]]$  code uses concatenation with 9 qubits and 8 stabiliser generators. The Steane  $[[7, 1, 3]]$  code uses CSS construction from the Hamming code, requiring only 7 qubits and 6 generators — 22% less overhead. The Steane code has cleanly separated X-type and Z-type stabilisers, while Shor's reflect concatenation hierarchy. Both support transversal CNOT, but Steane additionally supports transversal Hadamard due to the Hamming code's weak self-duality. Refer to sections 5.1.3 and 5.3.3 for details.
3.  $Y_5 = iX_5Z_5$  combines bit-flip and phase-flip on qubit 5. Z-type syndrome (detecting  $X_5$ ):  $g_4$  gives 1,  $g_5$  gives 0,  $g_6$  gives 1 — syndrome (1, 0, 1) identifies qubit 5. Apply  $X_5$  correction. X-type syndrome (detecting  $Z_5$ ):  $g_1$  gives 1,  $g_2$  gives 0,  $g_3$  gives 1 — syndrome (1, 0, 1) again identifies qubit 5. Apply  $Z_5$  correction. Combined  $Z_5X_5$  reverses the  $Y_5$  error, restoring the encoded state.
4. CSS codes separate generators into pure X-type and Z-type, simplifying fault-tolerant design. Syndrome circuits share identical CNOT structure with optional Hadamard wrapping for X-type. Transversal CNOT preserves CSS structure: X errors propagate forward, Z errors backward, maintaining independent correction. For self-dual CSS codes, transversal Hadamard swaps X-type and Z-type stabilisers. General codes mix X, Y, Z operators, complicating error propagation analysis. Refer to section 5.3.3 for detailed discussion.
5. The Steane  $[[7, 1, 3]]$  code uses 7 qubits with uncorrectable error probability  $\approx 0.84\%$  at 2% physical error rate. The  $[[11, 1, 5]]$  code uses 11 qubits with  $\approx 0.13\%$



uncorrectable error probability —  $6\times$  lower logical error rate. However, the  $[[11, 1, 5]]$  code lacks transversal Hadamard, requiring costly magic-state distillation. Recommendation: use the Steane code unless the application specifically demands multi-qubit correction and rarely uses Hadamard gates. Refer to section 5.3.3 for transversal gate details.

## 5.10 REFERENCES

### BOOKS

- Nielsen, M. A., & Chuang, I. L. (2010). *Quantum Computation and Quantum Information* (10th Anniversary ed.). Cambridge University Press.
- Gottesman, D. (1997). *Stabilizer Codes and Quantum Error Correction*. PhD Thesis, California Institute of Technology.

### JOURNAL ARTICLES

- Calderbank, A. R., & Shor, P. W. (1996). Good quantum error-correcting codes exist. *Physical Review A*, 54(2), 1098-1105.
- Steane, A. M. (1996). Error correcting codes in quantum theory. *Physical Review Letters*, 77(5), 793-797.

### WEB REFERENCES

- Roffe, J. (2019). Quantum Error Correction: An Introductory Guide. *Contemporary Physics*, 60(3), 226-245. <https://arxiv.org/abs/1910.03672>

Minor mathematical notation refinements (normalisation, exponent formatting, logical state representation) and stabiliser-measurement clarifications have been incorporated. No conceptual modifications were required.